

Multiple Choice (26 marks)

1. Which of the following facts was established prior to 1859, the year in which Charles Darwin published *On the Origin of Species*?
 - (a) DNA provides the macromolecular basis of heredity.
 - (b) Mendelian principles explain why some traits are dominant and others are recessive.
 - (c) Prokaryotes include two major domains, the Bacteria and the Archaea.
 - (d) There exist fossilized remains of species that have become extinct.
2. Which of the following would increase the rate at which a gas diffuses between the alveoli of the lung and the blood within a pulmonary capillary?
 - (a) Decreasing the partial pressure gradient of the gas
 - (b) Decreasing the solubility of the gas in water
 - (c) Increasing the total surface area available for diffusion
 - (d) Decreasing the rate of blood flow through the pulmonary capillary
3. Short-term changes in plant growth rate mediated by the plant hormone auxin are hypothesized to result from
 - (a) loss of turgor pressure in the affected cells
 - (b) increased extensibility of the walls of affected cells
 - (c) suppression of metabolic activity in affected cells
 - (d) cytoskeletal rearrangements in the affected cells
4. Targeting of a newly synthesized protein is most likely to require two different signal peptides for which of the following destinations?
 - (a) Plasma membrane
 - (b) Lysosome
 - (c) Cytosol
 - (d) Chloroplast
5. Charles Darwin's proposed conditions for natural selection encompass all of the following with regard to a given population EXCEPT
 - (a) inheritance of both "fit" and "unfit" genes
 - (b) differential survival and reproductive success

- (c) competition for limited resources
 - (d) overproduction of offspring
6. The first point of entry of water at the roots of a monocot plant is through the cytoplasm of cells of the
- (a) root cap
 - (b) Casparian strip
 - (c) pericycle
 - (d) endoderm
7. All of the following might be found in connective tissue EXCEPT
- (a) thrombin
 - (b) glycosaminoglycans
 - (c) collagens
 - (d) fibroblasts
8. Which of the following sets of reactions occurs in the stroma of the chloroplast in plant cells?
- (a) Calvin cycle
 - (b) Krebs cycle
 - (c) Fermentation
 - (d) Decarboxylation
9. Which of the following most accurately describes a retrotransposon?
- (a) A DNA sequence that can move from one site in the genome to another without replicating
 - (b) A DNA sequence that can be deleted from the genome without consequence
 - (c) A DNA sequence that replicates via an RNA intermediate
 - (d) A DNA sequence that replicates via a protein intermediate
10. Based on the characteristic population curves that result from plotting population growth of a species, the most effective means of controlling the mosquito...
- (a) maintain the population at a point corresponding to the midpoint of its logistic curve
 - (b) opt for zero population control once the K value of the curve has been reached
 - (c) reduce the carrying capacity of the environment to lower the K value
 - (d) increase the mortality rate

11. All of the following statements about plant embryogenesis are correct EXCEPT:
- (a) The suspensor is derived from the basal cell.
 - (b) Cotyledons are derived from the apical cell.
 - (c) Shoot apical meristem formation occurs after seed formation.
 - (d) Precursors of all three plant tissue systems are formed during embryogenesis.
12. The sight organs of crustaceans and insects contain ommatidia, which make up the individual visual units of the
- (a) eyespot
 - (b) simple eye
 - (c) compound eye
 - (d) binocular eye
13. Annelids and arthropods are similar to each other in that members of both phyla
- (a) have segmented bodies
 - (b) have a closed circulatory system
 - (c) conduct gas exchange by diffusion through a moist membrane
 - (d) have well-developed sense organs

True / False (4 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Keystone species are thought to have profound effects on the structure and composition of ecological communities because they' is 'tend to reduce diversity by eliminating food resources for other species'.
2. True or False: The correct answer to 'Which of the following depicts the correct sequence of membranes of the chloroplast, beginning with the innermost membrane and ending...' is 'Thylakoid membrane, inner membrane, outer membrane'.

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